STATISTICAL COMPUTING AND R PROGRAMMING LABORATORY

Analysis of Student GPA and Social Media Usage

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OVERVIEW:

This program explores the relationship between students' GPA and social media usage by leveraging statistical tools, forecasting models, and data visualization techniques. The menu-driven interface allows users to generate summary statistics, conduct ANOVA tests, create various plots, and make forecasts using ARIMA models. The analysis identifies significant effects of social media usage on GPA while testing specific hypotheses about usage patterns.

KEY FEATURES:

- **Menu-Driven Interface:** Users can choose from multiple actions, such as generating summary statistics, conducting ANOVA, performing hypothesis testing, and generating various plots through an intuitive menu.
- **Manual ANOVA Analysis:** Calculates between-group and within-group variance to assess the effect of social media usage categories (Low, Moderate, High) on GPA.
- **Hypothesis Testing:** Provides detailed t-tests to compare GPA differences between high and low usage groups and to analyze whether moderate usage results in a higher GPA than high usage.
- **Forecasting Using ARIMA:** Utilizes time series forecasting to predict future trends in GPA, offering improved insights into student performance.
- **Visualization Tools:** Includes bar plots, box plots, histograms, and time series plots to visually explore data distribution and trends.

This project offers a comprehensive approach to analyzing student performance in relation to social media behavior, providing valuable insights for educators and students to effectively balance screen time. The combination of statistical rigor, forecasting, and visualization enables both exploration of data and actionable decision-making.